A Rate–Work–Distortion Region for Consumer-Relative Observation

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Abstract

Source coding counts the bits needed to serve a consumer. A cyclic physical observer must also erase the memory that carried those bits. The two costs coincide only when no correlated side information survives at reset. We study a memory that stores a lossy description of an i.i.d. source, supports a declared consumer distortion without access to side information, and is later erased while correlated side information is retained. Under the classical, isothermal Landauer model, we characterize the complete achievable region of description rate, ideal reset work, and consumer distortion. For a test channel from the source to its reconstruction, the two coordinates are the ordinary mutual information and the mutual information conditioned on the reset side information. The work-only endpoint is the common-reconstruction rate–distortion function, but the full two-resource region distinguishes codes that have the same rate and distortion and different thermodynamic costs. We derive the supporting-channel equations for its Pareto boundary and give an exact binary example with a nontrivial rate–work frontier. We then prove a materialize-then-project barrier: once a full digital source record has been created, jointly resetting it and every derived description costs its full conditional entropy, regardless of the consumer's lower-dimensional need. We extend the region to several consumers, identify the conditional total correlation saved by coordinated reset, derive the Landauer scaling of the Gaussian read-operator water-filling solution, and obtain a temperature-weighted water-filling rule for heterogeneous memory banks. With jointly Gaussian reset side information, the scalar region collapses to a single corner whose rate–work gap grows to the full source–side-information mutual information as distortion vanishes, while the vector frontier reappears as a generalized water-filling in which the minimum-rate and minimum-work allocations differ. Finally, for a stale Markov record, predictive information lost with age is gained exactly as conditional erasure work. The results concern the reversible lower bound on reset work, not the total energy of sensing, communication, or finite-time operation.

Index Terms

Common reconstruction, consumer-relative distortion, information thermodynamics, Landauer principle, rate–distortion theory, semantic source coding, side information, source coding, thermodynamics of information.

I. Introduction

Source coding asks how many bits a consumer needs. A cyclic physical observer has a second obligation: it must eventually clear the memory that carried those bits. Landauer's principle assigns a thermodynamic cost to this logically irreversible step [1], [2]. In the ideal classical model, erasing one unbiased bit from a symmetric memory coupled to a bath at temperature T requires at least $k_{\mathrm{B}}T\ln 2$ of work. If correlated side information remains available, the relevant quantity is conditional entropy rather than entropy [3], [4]. Thus a description can be expensive to communicate and cheap to erase, or the reverse.

This paper studies that separation. A source block X^n is encoded into a memory index M. A consumer reconstructs $\widehat{X}^n$ from M alone and is judged by a declared distortion $d_{\mathcal{C}}$. Later, when M is reset, a correlated block S^n is available to the reset mechanism. The consumer never sees S^n. Figure 1 shows the distinction. The rate of the description is controlled by the size of M; its ideal reset work is controlled by $H(M \mid S^n)$.

The distortion is consumer-relative. Let $\mathcal{C}$ be the map that reads an observation and let $D_{\mathcal{Y}}$ be the loss on its output. The induced distortion may be written

$$d_{\mathcal{C}}(x,\hat{x}) = D_{\mathcal{Y}}(\mathcal{C}(x),\mathcal{C}(\hat{x}))\,. \tag{1}$$

This is the exact form used in Observation Theory [5]. Ordinary reconstruction is the special case in which $\mathcal{C}$ is the identity and $D_{\mathcal{Y}}$ is a source-space metric. The present paper takes $d_{\mathcal{C}}$ as given and asks a different question: what thermodynamic work is forced by a representation that meets it?

The main result is a rate–work–distortion region. For each test channel $p(\hat{x} \mid x)$ meeting the distortion constraint, the description-rate coordinate is $I(X;\widehat{X})$ and the normalized reset-work coordinate is $I(X;\widehat{X} \mid S)$. Their separate minima do not, in general, describe the full frontier: two test channels can have the same rate and distortion but different conditional work. The work-only endpoint coincides mathematically with the common-reconstruction rate–distortion function of Steinberg [6]. That connection is useful rather than accidental. In the common-reconstruction problem, side information reduces a communication rate while the reconstruction remains reproducible without it. Here the same conditional mutual information measures work saved when side information is available only at erasure. The full region keeps the ordinary description rate and the conditional reset

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This manuscript develops the thermodynamic extension of the author's Observation Theory program. The companion manuscript and reproducibility materials are available at https://github.com/ahb-sjsu/geometric-observation.

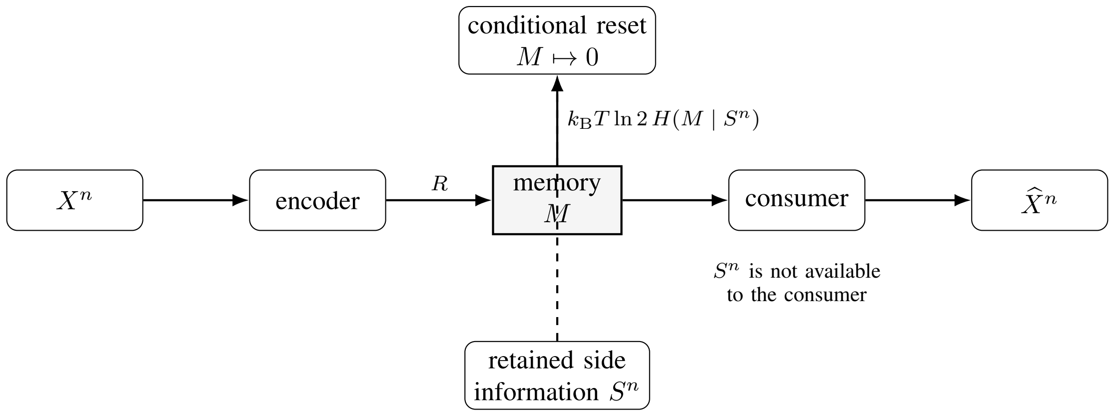

Fig. 1. The model. The consumer reconstructs from the stored description alone. Correlated side information is available only when the description memory is reset. Rate and reset work are therefore different resources.

work as distinct coordinates. This distinction is also related to the classical separation between fixed-rate and entropy-constrained quantization [7], [8]; the new ingredient is that the entropy coordinate is conditional on information retained by the reset mechanism and is assigned a thermodynamic meaning.

Six consequences follow. First, the lower boundary has a direct supporting-channel characterization, and even a two-bit source exhibits a strict tradeoff between description capacity and reset work. Second, a full digital record cannot be made thermodynamically cheap by compressing it after the fact. If a register holding X^n has already been materialized, the ideal work required to reset that register and every deterministic description derived from it is $k_{\mathrm{B}}T \ln 2\, H(X^n \mid S^n)$. Consumer-native observation saves work only when irrelevant distinctions are never committed to volatile memory, or are reversibly uncomputed. Third, a shared observation for several consumers is priced by the joint consumer output; coordinated reset saves their conditional total correlation. Fourth, for a Gaussian source and the read operator $P_{\mathcal{C}}$ of Observation Theory, the usual spectral water-filling rate becomes an isothermal work law. If eigenmodes are erased in memory banks at different temperatures, the water level is proportional to temperature. Fifth, with jointly Gaussian reset side information the scalar region is a single corner whose rate–work gap grows to $I(X;S)$ as distortion vanishes, and the tradeoff reappears in the vector case through the allocation of distortion across modes that the reset context knows unequally well. Sixth, for a stale Markov record, the loss of predictive information and the rise of conditional reset work are two sides of the same entropy identity.

The physical scope is deliberately narrow. We count the reversible lower bound for resetting a classical, degenerate memory in the asymptotic i.i.d. regime. We do not claim that sensing, transmitting, or switching a bit costs exactly $k_{\mathrm{B}}T \ln 2$. Real devices operate far above the Landauer scale, finite-time operation adds dissipation, and nondegenerate or quantum memories require more general free-energy and one-shot treatments [4], [9], [10]. The point is not a present-day power model. It is to identify which distinctions must be physically instantiated before engineering losses are added.

II. Model and physical accounting

All logarithms are base two. Random variables have finite alphabets unless a Gaussian specialization is stated. Let $(X,S) \sim p_{XS}$, and let (X^n, S^n) be i.i.d. copies. The encoder observes X^n but not S^n. The consumer and its decoder also do not observe S^n. The reset mechanism alone has access to S^n, which remains unchanged by the reset.

A consumer-relative distortion is a bounded function

$$d_{\mathcal{C}} : \mathcal{X} \times \widehat{\mathcal{X}} \longrightarrow [0, d_{\max}]. \tag{2}$$

It may be a conventional fidelity criterion or the induced loss in (1). An $(n, \mathcal{M}_n)$ observation code consists of an encoder and decoder

$$f_n : \mathcal{X}^n \to \mathcal{M}_n, \qquad g_n : \mathcal{M}_n \to \widehat{\mathcal{X}}^n. \tag{3}$$

We write $M = f_n(X^n)$ and $\widehat{X}^n = g_n(M)$. Randomized encoders do not improve the region below, but allowing them in the converse is harmless.

The three operational quantities are

$$R_n = \frac{1}{n} \log |\mathcal{M}_n|, \tag{4}$$

$$D_n = \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}\, d_{\mathcal{C}}(X_i, \widehat{X}_i), \tag{5}$$

$$L_n = \frac{1}{n} H(M \mid S^n). \tag{6}$$

The dimensionless quantity L_n is the *Landauer content* of the memory relative to the retained side information. Under an ideal conditional reset, the average work satisfies

$$W_n \;\geq\; nk_{\mathrm{B}}T\ln 2\, L_n. \tag{7}$$

In the classical asymptotic regime, conditional erasure protocols make this bound tight up to sublinear terms [3], [4]. We therefore state the coding theorem in bits per source symbol and restore physical units by multiplying by $k_{\mathrm{B}}T\ln 2$.

Definition 1 (Achievability)**.** A triple (R, L, D) is achievable if there is a sequence of observation codes such that

$$\limsup_{n\to\infty} R_n \leq R, \qquad \limsup_{n\to\infty} L_n \leq L, \qquad \limsup_{n\to\infty} D_n \leq D. \tag{8}$$

The closure of the achievable set at distortion D is denoted $\mathcal{RW}_{\mathcal{C}}(D)$.

The codebook, wiring, and any permanent lookup tables are treated as part of the apparatus rather than a per-cycle record. Only the volatile index M is reset. This is the same separation made in ordinary source coding between the code and the message produced on one source block.

Remark 1 (What the model does not count). Equation (7) is not a universal energy charge per observed or transmitted bit. It does not include the work of acquiring X^n, driving a channel, maintaining metastable states, correcting errors, or finishing in finite time. It prices the logically irreversible reset of the memory state that the code actually produced. This distinction is essential: reversible computation can retain and uncompute intermediate information [2], whereas an overwritten record has been discarded.

Remark 2 (Semantic invariance of erasure). For a fixed joint state p_{MS^n}, fixed memory physics, temperature, and reset operation, the minimum reset work is independent of the identity, purpose, or loss function of any downstream consumer. The consumer enters this paper only by constraining the set of admissible codes through the distortion requirement. The causal chain is

$$\text{consumer requirement} \longrightarrow \text{code design} \longrightarrow p_{MS^n} \longrightarrow \text{reset work},$$

and once the code and the physical state are fixed, the consumer drops out of the thermodynamic accounting. Two senses of "observer" must accordingly be kept distinct. The Observation-Theory *consumer* is a downstream functional that defines the operative distortion. The *reset agent* of conditional-erasure thermodynamics is a physical system holding the correlated state S^n that participates in the erasure protocol [3]. Theorem 2 is the sharp form of this locality: a record already materialized cannot be made cheaper by a later declaration of irrelevance.

III. The rate–work–distortion region

For a test channel $p(\hat{x} \mid x)$, the Markov chain $\widehat{X} - X - S$ holds because side information is unavailable when the observation is formed. Define

$$\mathcal{T}_{\mathcal{C}}(D) = \left\{ p(\hat{x} \mid x) : \mathbb{E} d_{\mathcal{C}}(X, \widehat{X}) \leq D \right\}. \tag{9}$$

Lemma 1 (Convexity in the test channel)**.** *For fixed p_{XS}, both coordinates $I(X; \widehat{X})$ and $I(X; \widehat{X} \mid S)$ are convex functions of the test channel $q(\hat{x} \mid x)$.*

Proof. Convexity of $I(X; \widehat{X})$ in the channel at fixed input is classical. For the conditional coordinate, the Markov chain $\widehat{X} - X - S$ gives $p(\hat{x} \mid x, s) = q(\hat{x} \mid x)$, so

$$I(X; \widehat{X} \mid S) = \sum_s p(s)\, I_{p(x|s)}(q), \tag{10}$$

a nonnegative combination of mutual informations, each with fixed input distribution $p(x \mid s)$ and the common channel q; each term is convex in q, hence so is the sum. We note that the decomposition $I(X; \widehat{X} \mid S) = I(X; \widehat{X}) - I(S; \widehat{X})$ does *not* yield this conclusion: $I(S; \widehat{X})$ is itself convex in q (the induced channel $p(\hat{x} \mid s) = \sum_x p(x \mid s) q(\hat{x} \mid x)$ is linear in q), so the difference is a priori indefinite. $\square$

Theorem 1 (Rate–work–distortion region)**.** *For a discrete memoryless source (X, S) and bounded consumer distortion $d_{\mathcal{C}}$,*

$$\boxed{\mathcal{RW}_{\mathcal{C}}(D) = \bigcup_{p(\hat{x}|x) \in \mathcal{T}_{\mathcal{C}}(D)} \left\{ (R, L) : R \geq I(X; \widehat{X}),\ L \geq I(X; \widehat{X} \mid S) \right\}.} \tag{11}$$

The physical reset-work coordinate is $W/n = k_{\mathrm{B}}T\ln 2\, L$ in the ideal model. By Lemma 1 the union is already convex, and it is closed by compactness of $\mathcal{T}_{\mathcal{C}}(D)$ and continuity of both coordinates; no time sharing is required, and every boundary point is attained by a single test channel.

Proof. We first prove the converse. Let a code produce M and $\widehat{X}^n$. Since $\widehat{X}^n$ is a function of M,

$$\begin{aligned}\log|\mathcal{M}_n| &\geq H(M) \geq I(X^n;M) \geq I(X^n;\widehat{X}^n)\\ &\geq \sum_{i=1}^{n} I(X_i;\widehat{X}_i),\end{aligned} \tag{12}$$

where the last step follows from memorylessness and conditioning reducing entropy. Likewise,

$$\begin{aligned}H(M \mid S^n) &\geq I(X^n;M \mid S^n) \geq I(X^n;\widehat{X}^n \mid S^n)\\ &\geq \sum_{i=1}^{n} I(X_i;\widehat{X}_i \mid S_i).\end{aligned} \tag{13}$$

Each induced per-letter channel $p(\hat{x}_i \mid x_i)$ satisfies the Markov constraint $\widehat{X}_i - X_i - S_i$: $\widehat{X}_i$ is a function of $(X_i, X_{\neq i})$ and S_i is independent of $X_{\neq i}$ given X_i for i.i.d. pairs. Let $\bar{q}(\hat{x} \mid x) = \frac{1}{n}\sum_{i=1}^{n} p(\hat{x}_i = \hat{x} \mid X_i = x)$ be the mixture channel; the constraint survives the mixture because $p(s \mid x)$ is common across i. Its distortion is D_n, so $\bar{q} \in \mathcal{T}_{\mathcal{C}}(D_n)$, and by Lemma 1,

$$R_n \;\geq\; \frac{1}{n}\sum_i I(X_i;\widehat{X}_i) \;\geq\; I_{\bar{q}}(X;\widehat{X}), \qquad L_n \;\geq\; \frac{1}{n}\sum_i I(X_i;\widehat{X}_i \mid S_i) \;\geq\; I_{\bar{q}}(X;\widehat{X} \mid S),$$

which places (R_n, L_n) in the (already convex and closed) union of (11) at distortion D_n.

For achievability, fix $p(\hat{x} \mid x) \in \mathcal{T}_{\mathcal{C}}(D)$ and $\epsilon > 0$. Generate a standard rate–distortion codebook containing $2^{n(I(X;\widehat{X})+2\epsilon)}$ independent $\hat{x}^n$ codewords. A covering encoder selects a codeword jointly typical with x^n and stores its index M. The consumer reconstructs that codeword from M alone. Standard typicality arguments give rate $I(X;\widehat{X})+2\epsilon$ and distortion at most $D+o(1)$.

To bound the reset work, randomly partition the codeword indices into $2^{n(I(X;\widehat{X}|S)+3\epsilon)}$ bins, and let $B = b(M)$ be the bin index. Because $\widehat{X} - X - S$,

$$I(X;\widehat{X} \mid S) = I(X;\widehat{X}) - I(S;\widehat{X}). \tag{14}$$

The usual Wyner–Ziv packing argument therefore gives a decoder that recovers M from (B, S^n) with probability of error tending to zero. This decoder is a proof device; the consumer still receives no side information. By Fano's inequality,

$$\begin{aligned}H(M \mid S^n) &\leq H(B) + H(M \mid B, S^n)\\ &\leq n\big(I(X;\widehat{X} \mid S) + 3\epsilon\big) + o(n).\end{aligned} \tag{15}$$

Thus the same stored index has ordinary description rate $I(X;\widehat{X})$ and conditional Landauer content $I(X;\widehat{X} \mid S)$. Since the union in (11) is already convex and closed (Lemma 1), this exhibits the entire region. ☐

The binning step in the proof has a simple interpretation. The full codeword index is required by the consumer. At reset, side information identifies the codeword up to a bin of asymptotic size $2^{nI(X;\widehat{X}|S)}$. Only that residual uncertainty is thermodynamically irreducible. The step is also operationally checkable at finite blocklength. In the reproducibility materials, a random codebook on the source of Proposition 2 stores an index whose in-bin maximum-likelihood recovery from (B, S^n) succeeds at bin rates near 0.39 of the measured description rate and fails increasingly below the measured conditional content, while the same decoder without S^n fails at every bin rate tested.

A. Endpoints and relation to common reconstruction

Define the ordinary consumer-relative rate–distortion function

$$R_{\mathcal{C}}(D) = \min_{p(\hat{x}|x)\in\mathcal{T}_{\mathcal{C}}(D)} I(X;\widehat{X}) \tag{16}$$

and the minimum Landauer content

$$L_{\mathcal{C}}(D \mid S) = \min_{p(\hat{x}|x)\in\mathcal{T}_{\mathcal{C}}(D)} I(X;\widehat{X} \mid S). \tag{17}$$

The two minima can be attained by different test channels, which is why the region in Theorem 1 contains more information than its two axis intercepts.

Corollary 1 (Work endpoint)**.** *The minimum ideal reset work per source symbol is*

$$W^{\star}_{\mathcal{C}}(D, T \mid S) = k_{\mathrm{B}} T \ln 2\, L_{\mathcal{C}}(D \mid S). \tag{18}$$

The information quantity $L_{\mathcal{C}}(D \mid S)$ is Steinberg's common-reconstruction rate–distortion function for decoder side information S [6]. If $R_{\mathrm{WZ},\mathcal{C}}(D \mid S)$ denotes the Wyner–Ziv function, then

$$R_{\mathrm{WZ},\mathcal{C}}(D \mid S) \leq L_{\mathcal{C}}(D \mid S) \leq R_{\mathcal{C}}(D). \tag{19}$$

Proof. The first statement is Theorem 1 with rate unconstrained. Common reconstruction requires the reproduction to be determined without using S, which gives (17). Wyner–Ziv reconstruction may depend on S and therefore has a larger feasible set. The second inequality follows by conditioning: $I(X;\widehat{X} \mid S) \le I(X;\widehat{X})$ for every admissible test channel under $\widehat{X} - X - S$. □

The corollary is also a novelty boundary. The conditional mutual information in (17) is not a new source-coding functional. What is new in the present formulation is its role as one coordinate of a joint rate–work–distortion region when the full description, not merely a bin, must remain available to a consumer that lacks S.

B. Supporting channels on the lower boundary

Because the region is convex, each exposed point of its lower boundary minimizes a weighted sum of description rate and Landauer content. The resulting channel equation generalizes the Blahut–Arimoto fixed point.

Proposition 1 (Pareto-channel equation). *Fix $\alpha \in [0,1]$ and a distortion multiplier $\beta \ge 0$. Let $q(\hat{x} \mid x)$ minimize*

$$J_{\alpha,\beta}(q) = \alpha I(X;\widehat{X}) + (1-\alpha) I(X;\widehat{X} \mid S) + \beta \mathbb{E} d_{\mathcal{C}}(X,\widehat{X}). \tag{20}$$

On any support on which $q(\hat{x} \mid x) > 0$, the minimizing channel satisfies

$$\boxed{q(\hat{x} \mid x) = \frac{q(\hat{x})^{\alpha} \prod_s q(\hat{x} \mid s)^{(1-\alpha)p(s|x)} 2^{-\beta d_{\mathcal{C}}(x,\hat{x})}}{\sum_{\tilde{x}} q(\tilde{x})^{\alpha} \prod_s q(\tilde{x} \mid s)^{(1-\alpha)p(s|x)} 2^{-\beta d_{\mathcal{C}}(x,\tilde{x})}}.} \tag{21}$$

Here $q(\hat{x}) = \sum_x p(x) q(\hat{x} \mid x)$ and $q(\hat{x} \mid s) = \sum_x p(x \mid s) q(\hat{x} \mid x)$. Conversely, an interior channel satisfying (21) and the KKT conditions is globally optimal for (20).

Proof. By Lemma 1, both mutual informations in (20) are convex in the test channel for fixed p_{XS}, and the distortion term is linear. The KKT conditions are therefore sufficient. Differentiating with respect to $q(\hat{x} \mid x)$, imposing one normalization multiplier per x, and collecting the coefficient of $\log_2 q(\hat{x} \mid x)$ gives

$$\begin{aligned}\log_2 q(\hat{x} \mid x) = \alpha \log_2 q(\hat{x}) + (1-\alpha) \sum_s p(s \mid x) \log_2 q(\hat{x} \mid s) \\ - \beta d_{\mathcal{C}}(x,\hat{x}) - \kappa(x).\end{aligned} \tag{22}$$

Exponentiation and normalization over $\hat{x}$ yield (21). □

For $\alpha = 1$, (21) is the ordinary rate–distortion fixed point. For $\alpha = 0$, it supports the minimum-work, or common-reconstruction, endpoint. Intermediate values expose the rate–work tradeoff rather than either intercept alone.

Corollary 2 (Exact consumer output). *Let $U = \mathcal{C}(X)$ be a deterministic consumer output, and suppose zero distortion means exact recovery of U but not necessarily of X. Then*

$$R_{\mathcal{C}}(0) = H(U), \qquad L_{\mathcal{C}}(0 \mid S) = H(U \mid S). \tag{23}$$

Thus the exact thermodynamic object is the consumer quotient U, not the full source state.

Proof. Choose the reproduction to be U. Conversely, any zero-distortion reproduction determines U, so data processing gives the matching lower bounds. □

C. An exact binary rate–work frontier

The smallest nontrivial example already separates the two resources.

Proposition 2 (Two independent bits). *Let $X = (A,B)$, where A and B are independent fair bits, and let the reset side information be $S = A$. The consumer reconstructs both components under average Hamming distortion*

$$d(x,\hat{x}) = \tfrac{1}{2}\mathbf{1}\{a \ne \hat{a}\} + \tfrac{1}{2}\mathbf{1}\{b \ne \hat{b}\}. \tag{24}$$

For $0 \le D \le 1/4$, the lower Pareto boundary of $\mathcal{RW}_{\mathcal{C}}(D)$ is parameterized by $0 \le t \le D$ as

$$D_A = t, \qquad D_B = 2D - t, \tag{25}$$
$$R(t) = 2 - h_2(t) - h_2(2D-t), \tag{26}$$
$$L(t) = 1 - h_2(2D-t). \tag{27}$$

It is achieved by independent binary symmetric test channels on A and B.

Proof. For an arbitrary test channel with reproduction $Y = (\widehat{A}, \widehat{B})$, let $D_A = \Pr\{A \neq \widehat{A}\}$ and $D_B = \Pr\{B \neq \widehat{B}\}$; its region coordinates satisfy, since A and B are independent,

$$\begin{aligned} R &= I(A, B; Y) \\ &= I(A; Y) + I(B; Y \mid A) \\ &\geq 1 - h_2(D_A) + 1 - h_2(D_B), \end{aligned} \tag{28}$$

$$\begin{aligned} L &= I(A, B; Y \mid A) = I(B; Y \mid A) \\ &\geq 1 - h_2(D_B). \end{aligned} \tag{29}$$

The first bound uses data processing for $\widehat{A}$ and the binary rate–distortion converse for B conditional on A; convexity of the binary rate–distortion function combines the conditional distortions. The second is the same conditional converse. Independent binary symmetric test channels meet both bounds simultaneously.

An undominated point uses the full distortion budget, so $D_A + D_B = 2D$. On $0 \leq t = D_A \leq D$, $R(t)$ decreases while $L(t)$ increases. For $D < t \leq 2D$, the reflected allocation $2D - t$ has the same rate and strictly smaller work, so that half of the curve is dominated. Equations (26)–(27) follow. □

Figure 2 shows the frontier at $D = 0.15$. Its endpoints have clear meanings. At $t = 0$, all allowed error is placed on B, the component unknown at reset; this minimizes work but requires more description capacity. At $t = D$, distortion is divided evenly; this minimizes rate but leaves more uncertainty to erase. No single scalar "number of bits" describes both choices.

The matched-rate inversion is immediate. The allocations $(D_A, D_B) = (0, 0.30)$ and $(0.30, 0)$ both have

$$R = 2 - h_2(0.30) = 1.1187 \text{ bits/source symbol} \tag{30}$$

and total distortion 0.15. Their Landauer contents are respectively 0.1187 and 1 bit/source symbol. The second point is not Pareto optimal, but it isolates the physical effect: two descriptions tied by ordinary rate and distortion can differ sharply in reset work.

IV. The Materialize-then-Project Barrier

Theorem 1 prices the memory index actually produced by the observer. It does not say that already recorded information becomes free when a later computation declares it irrelevant. The next result makes that distinction exact.

Theorem 2 (Materialization barrier)**.** *Suppose a volatile register A first stores the full discrete source block, $A = X^n$, and a second volatile register stores a deterministic description $M = f_n(A)$. If both registers are returned to a standard state while S^n is retained, then their minimum asymptotic ideal reset work is*

$$W^{\star}_{A,M} = k_{\mathrm{B}} T \ln 2\, H(A, M \mid S^n) = n k_{\mathrm{B}} T \ln 2\, H(X \mid S). \tag{31}$$

This cost is independent of the distortion met by M.

Proof. Conditional erasure of the pair (A, M) costs its conditional entropy in the ideal model. Since $A = X^n$ and M is a function of A,

$$H(A, M \mid S^n) = H(A \mid S^n) = H(X^n \mid S^n) = nH(X \mid S). \tag{32}$$

The same result appears under sequential optimal reset. For example,

$$H(A \mid M, S^n) + H(M \mid S^n) = H(A, M \mid S^n). \tag{33}$$

The identity is the chain rule. □

The theorem does not assign a universal work cost to measurement. It says something narrower: once a complete digital record has been committed to a volatile logical state, projecting it later cannot refund the work needed to clear that record. One may retain the discarded detail as reversible garbage and uncompute it, or avoid recording it. Overwriting it is the irreversible step.

For a direct consumer-native observer, only M is materialized. Combining Theorem 2 with Corollary 1 gives the ideal work gap

$$\Delta W_{\mathcal{C}}(D, T \mid S) = k_{\mathrm{B}} T \ln 2 \left[H(X \mid S) - L_{\mathcal{C}}(D \mid S) \right]. \tag{34}$$

At zero distortion for a deterministic consumer output $U = \mathcal{C}(X)$,

$$\Delta W_{\mathcal{C}}(0, T \mid S) = k_{\mathrm{B}} T \ln 2\, H(X \mid U, S). \tag{35}$$

The conditional entropy in (35) is the physical price of materializing distinctions that neither the consumer nor the retained reset context needs.

This result has an architectural reading. Post-hoc semantic compression can save communication and downstream storage. It cannot recover the Landauer saving if a full volatile source record was already created and must be erased. The thermodynamic advantage requires projection at or before the first digital memory boundary.

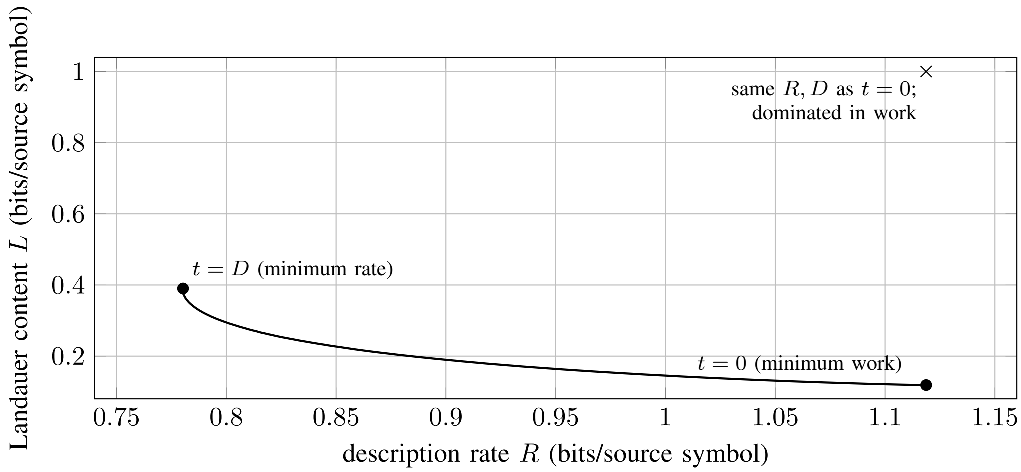

Fig. 2. Exact lower rate–work frontier for Proposition 2 at distortion $D = 0.15$. The marked point at $L = 1$ comes from reversing the two component distortions: it has the same rate and source distortion as the minimum-work endpoint, but about 8.4 times the Landauer content.

V. Several consumers

A practical observer often serves several consumers: a controller, an alarm, an auditor, and a forensic process may read different functions of the same source. Let consumer j receive $\widehat{X}_j^n = g_{j,n}(M)$ and incur distortion $d_j(X_i, \widehat{X}_{j,i})$.

Theorem 3 (Shared observation for several consumers)**.** *For distortion vector $\boldsymbol{D} = (D_1, \ldots, D_m)$, the achievable rate–work region of one shared memory is*

$$\begin{aligned}\mathcal{RW}_{1:m}(\boldsymbol{D}) = \overline{\operatorname{conv}} \bigcup_{p(\hat{x}_1,\ldots,\hat{x}_m|x)} \big\{(R,L) : R \geq I(X; \widehat{X}_1, \ldots, \widehat{X}_m),\\ L \geq I(X; \widehat{X}_1, \ldots, \widehat{X}_m \mid S)\big\},\end{aligned} \tag{36}$$

where the union is over joint test channels satisfying $\mathbb{E}d_j(X, \widehat{X}_j) \leq D_j$ for every j.

Proof. Treat the vector reproduction $\widehat{\boldsymbol{X}} = (\widehat{X}_1, \ldots, \widehat{X}_m)$ as the reproduction alphabet in Theorem 1. The covering step enforces all m distortion constraints simultaneously through joint typicality with the vector reproduction, and the converse applies verbatim with the vector constraint set in place of $\mathcal{T}_{\mathcal{C}}(D)$. Each consumer obtains its component by a deterministic projection. □

Corollary 3 (Exact consumers and coordinated reset)**.** *Let $U_j = \mathcal{C}_j(X)$ be exact deterministic consumer outputs. A joint memory, or separate memories reset with full cross-memory coordination, has ideal reset content*

$$H(U_1, \ldots, U_m \mid S). \tag{37}$$

If the m memories are reset independently and each reset sees only S, their contents sum to $\sum_j H(U_j \mid S)$. The excess is the conditional total correlation

$$\operatorname{TC}(U_1; \ldots; U_m \mid S) = \sum_{j=1}^{m} H(U_j \mid S) - H(U_1, \ldots, U_m \mid S) \geq 0. \tag{38}$$

The qualification about coordinated reset matters. The saving in (38) is not attached to a particular memory package. It is obtained whenever the erasure protocol can use the other consumer records as side information before they are cleared. Independent administrative or physical reset domains forfeit that correlation.

For observation architecture, the corollary gives a thermodynamic reason to use a common base representation with consumer-specific refinements. Shared consumer structure is not merely a compression opportunity; it is conditional entropy that need not be paid more than once at reset.

VI. Gaussian read geometry, reset side information, and thermal water-filling

The preceding results accept any bounded single-letter distortion. We now connect them to the local read geometry of Observation Theory. Let $X \sim \mathcal{N}(0, \Sigma_X)$ and let a smooth consumer have Jacobian $J_{\mathcal{C}}$ and output metric $G \succeq 0$. Its local read operator is

$$P_{\mathcal{C}} = J_{\mathcal{C}}^{\mathsf{T}} G J_{\mathcal{C}} \succeq 0, \tag{39}$$

and the quadratic consumer distortion is

$$d_{\mathcal{C}}(x,\hat{x}) = (x-\hat{x})^{\mathsf{T}} P_{\mathcal{C}}(x-\hat{x}). \tag{40}$$

For a nonlinear consumer, this is the second-order approximation to its output loss; for a linear consumer with quadratic output loss it is exact [5].

Let $\lambda_1,\ldots,\lambda_r > 0$ be the nonzero eigenvalues of $\Sigma_X^{1/2} P_{\mathcal{C}} \Sigma_X^{1/2}$, where $r = \operatorname{rank}(\Sigma_X^{1/2} P_{\mathcal{C}} \Sigma_X^{1/2})$. In an isothermal memory with no reset side information, rate and Landauer content coincide.

Corollary 4 (Isothermal read-operator work law)**.** *For $0 < D < \sum_{i=1}^r \lambda_i$, let θ be chosen so that*

$$D = \sum_{i=1}^{r} \min\{\lambda_i, \theta\}. \tag{41}$$

Then

$$R_{\mathcal{C}}(D) = L_{\mathcal{C}}(D) = \frac{1}{2}\sum_{i=1}^{r}\left[\log_2 \frac{\lambda_i}{\theta}\right]_+, \tag{42}$$

$$W^{\star}_{\mathcal{C}}(D,T) = \frac{k_{\mathrm{B}}T}{2}\sum_{i=1}^{r}\left[\ln \frac{\lambda_i}{\theta}\right]_+. \tag{43}$$

Directions in $\ker P_{\mathcal{C}}$ require neither rate nor reset work, provided they are never materialized in the observation memory.

Proof. Equation (42) is the weighted Gaussian rate–distortion solution obtained by diagonalizing the read covariance $\Sigma_X^{1/2} P_{\mathcal{C}} \Sigma_X^{1/2}$ and applying reverse water-filling. Multiply bits by $k_{\mathrm{B}}T\ln 2$ to obtain (43). □

The final proviso is the materialization barrier in spectral form. A null read-direction is free to omit. It is not free to erase after it has been stored.

Reset side information changes the Gaussian picture in a way the scalar case makes exact.

Proposition 3 (Scalar Gaussian corner)**.** *Let (X,S) be jointly Gaussian and zero mean with $\operatorname{Var} X = \sigma^2$ and correlation coefficient $\rho \in (-1,1)$, let $d(x,\hat{x}) = (x-\hat{x})^2$, and let $0 < D < \sigma^2$. Then*

$$\mathcal{RW}(D) = \left\{(R,L) : R \geq \tfrac{1}{2}\log_2 \tfrac{\sigma^2}{D},\ L \geq \tfrac{1}{2}\log_2 \tfrac{\sigma^2(1-\rho^2)+\rho^2 D}{D}\right\}. \tag{44}$$

Both bounds are attained simultaneously by the classical reverse channel, so the scalar Gaussian problem has no rate–work tradeoff. The side-information discount

$$R_{\min} - L_{\min} = \tfrac{1}{2}\log_2 \frac{\sigma^2}{\sigma^2(1-\rho^2)+\rho^2 D} \tag{45}$$

is decreasing in D and increases to $I(X;S)$ as $D \to 0$.

Proof. Normalize $\sigma = \sigma_S = 1$, and assume $\mathbb{E}\widehat{X} = 0$ without loss of generality; centering $\widehat{X}$ changes neither mutual information and does not increase the distortion. Under $\widehat{X} - X - S$, $\mathbb{E}[S\widehat{X}] = \mathbb{E}\big[\mathbb{E}[S \mid X]\,\mathbb{E}[\widehat{X} \mid X]\big] = \rho\,\mathbb{E}[X\widehat{X}]$, so with $c = \operatorname{Cov}(X,\widehat{X})$ and $v = \operatorname{Var}\widehat{X} \geq c^2$ the second moments of $(X,\widehat{X},S)$ are determined by (c,v), and the distortion constraint reads $1 - 2c + v \leq D$; together these force $c > 0$ on the feasible set. For the work coordinate, $L = h(X \mid S) - h(X \mid \widehat{X}, S)$ and $h(X \mid \widehat{X},S) \leq \frac{1}{2}\log 2\pi e\, e_{\mathrm{lin}}$, where e_{lin} is the error of the best linear estimate of X from $(\widehat{X},S)$; the bound holds for any reproduction alphabet, discrete included. A direct computation gives $e_{\mathrm{lin}} = (1-\rho^2)(v-c^2)/(v-\rho^2c^2)$, hence

$$L \;\geq\; \tfrac{1}{2}\log_2 \frac{v-\rho^2c^2}{v-c^2}. \tag{46}$$

The right side of (46) decreases in v, so the distortion constraint binds, $v = D - 1 + 2c$. On that line the stationarity condition of the ratio reduces to $(1-\rho^2)\,c\,(c+D-1) = 0$; the root $c = 0$ is infeasible, the interior point $c = 1-D$ has positive second derivative, the boundary values diverge, and the minimum is $(1-\rho^2+\rho^2D)/D$. The rate converse is classical. The reverse channel $\widehat{X} = (1-D)X + W$ with $W \sim \mathcal{N}(0, D(1-D))$ independent of (X,S) has $c = v = 1-D$, distortion exactly D, and attains both bounds, so the region is the stated quadrant. Monotonicity of (45) and the limit $\frac{1}{2}\log_2 \frac{1}{1-\rho^2} = I(X;S)$ are elementary. □

Within the jointly Gaussian family, L is in fact a function of R alone, $L = \frac{1}{2}\log_2\!\big(1 + (1-\rho^2)(2^{2R}-1)\big)$; a scalar source offers no allocation freedom. The binary frontier of Proposition 2 came from allocating distortion between a component the reset context knows and one it does not. The next proposition restores exactly that mechanism.

Proposition 4 (Vector frontier and reset water-filling)**.** *Let X have independent components $X_i \sim \mathcal{N}(0,\sigma_i^2)$, $i = 1,\ldots,r$, let $S = (S_1,\ldots,S_r)$ with S_i jointly Gaussian with X_i alone at correlation ρ_i, and let $d(x,\hat{x}) = \sum_i p_i(x_i-\hat{x}_i)^2$ with $p_i > 0$, the* *read geometry of* (40) *with independently read and independently known modes. Write* $c_i = \sigma_i^2(1-\rho_i^2)$*. The lower Pareto boundary of* $\mathcal{RW}(D)$ *is traced by* $\alpha \in [0,1]$ *as follows. With* $\mu > 0$ *chosen so that* $\sum_i p_i d_i = D$*, allocate* $d_i = \min\{\sigma_i^2, \delta_i(\mu)\}$*, where* $\delta_i(\mu)$ *is the positive root of*

$$2\mu p_i \rho_i^2\, \delta^2 + \left(2\mu p_i c_i - \alpha\rho_i^2\right)\delta - c_i = 0 \tag{47}$$

and $\delta_i = 1/(2\mu p_i)$ *when* $\rho_i = 0$*; the frontier point is*

$$R = \sum_{i=1}^{r} \tfrac{1}{2}\log_2 \frac{\sigma_i^2}{d_i}, \qquad L = \sum_{i=1}^{r} \tfrac{1}{2}\log_2 \frac{c_i + \rho_i^2 d_i}{d_i}. \tag{48}$$

At $\alpha = 1$ *the rule is classical reverse water-filling. At* $\alpha = 0$ *it allocates distortion toward the modes the reset context knows least. A mode with* $d_i = \sigma_i^2$ *contributes zero to both coordinates.*

Proof. For independent pairs (X_i, S_i) and any code with $\widehat{X} - X - S$, the componentwise analogue of Appendix A gives $I(X;\widehat{X}) \ge \sum_i I(X_i;\widehat{X}_i)$ and $I(X;\widehat{X} \mid S) \ge \sum_i I(X_i;\widehat{X}_i \mid S_i)$. Proposition 3 bounds each mode at its own distortion $d_i = \mathbb{E}(X_i - \widehat{X}_i)^2$, so every achievable pair is dominated by the allocation program that minimizes $\alpha\sum_i R_i(d_i) + (1-\alpha)\sum_i L_i(d_i)$ over $\sum_i p_i d_i \le D$, $0 < d_i \le \sigma_i^2$. Each R_i is convex in d_i, and

$$L_i''(d) \;\propto\; \frac{1}{d^2} - \frac{\rho_i^4}{(c_i + \rho_i^2 d)^2} \;>\; 0 \quad \text{exactly when } c_i > 0,$$

so the program is convex and its KKT condition

$$\frac{\alpha}{2d_i} + \frac{(1-\alpha)\,c_i}{2d_i\left(c_i + \rho_i^2 d_i\right)} = \mu p_i \tag{49}$$

is sufficient; clearing denominators gives (47). Products of scalar reverse channels achieve every allocation, convexity of the per-mode objectives makes the achievable set convex, and the weighted-sum sweep therefore traces the entire lower boundary; strict convexity makes the minimizer unique at every α, endpoints included. The last claim is the identity $c_i + \rho_i^2\sigma_i^2 = \sigma_i^2$. □

Setting every $\rho_i = 0$ gives $L \equiv R$ and recovers Corollary 4. The tradeoff lives entirely in the heterogeneity of the ρ_i and σ_i. For two unit-variance modes with $\rho = (0.95, 0)$, equal read weights, and $D = 0.5$, the minimum-rate and minimum-work allocations differ by about 0.11 bits of Landauer content and 0.13 bits of rate at matched distortion; the reproducibility materials carry the exact frontier. This is the Gaussian analogue of the binary frontier, with the water level of (49) playing the role of the allocation parameter t.

The Landauer scale also changes the allocation rule when different components of the observation are reset against different reservoirs. Consider physically separate memory banks, with eigenmode i reset at temperature $T_i > 0$. Let $d_i \le \lambda_i$ be its allocated distortion. Independent Gaussian coding requires $r_i = \frac{1}{2}\log_2(\lambda_i/d_i)$ bits for an active mode and costs $k_{\mathrm{B}} T_i \ln 2\, r_i$ of ideal reset work.

Proposition 5 (Temperature-weighted water-filling)**.** *Under the independent eigenmode memory model, the minimum ideal reset work at total distortion* D *is attained by*

$$d_i^\star = \min\{\lambda_i, \nu T_i\}, \qquad \sum_{i=1}^{r} d_i^\star = D, \tag{50}$$

for a unique $\nu > 0$ *in the nontrivial range. The minimum work is*

$$\boxed{W_{\text{het}}^\star(D) = \frac{k_{\mathrm{B}}}{2}\sum_{i=1}^{r} T_i \left[\ln \frac{\lambda_i}{\nu T_i}\right]_+ .} \tag{51}$$

When all T_i *are equal,* (50) *reduces to ordinary reverse water-filling.*

Proof. For active mode i,

$$W_i(d_i) = \frac{k_{\mathrm{B}} T_i}{2}\ln\frac{\lambda_i}{d_i}. \tag{52}$$

Minimize $\sum_i W_i(d_i)$ subject to $0 < d_i \le \lambda_i$ and $\sum_i d_i \le D$. The KKT stationarity condition for an active mode is

$$-\frac{k_{\mathrm{B}} T_i}{2d_i} + \mu = 0, \tag{53}$$

so $d_i = (k_{\mathrm{B}}/2\mu) T_i = \nu T_i$. Clipping at λ_i gives (50); substitution gives (51). □

The rule allocates more distortion to modes stored in hotter banks. This is not a claim about one isothermal register. It is a design result for a physically partitioned observer whose reset environments have different marginal work per bit.

VII. Staleness as lost relevance and increased reset work

Side information at reset need not be a static auxiliary variable. It may be a later state of the process that generated the record. This produces a simple link between age, relevance, and work.

Proposition 6 (Staleness-work complement)**.** *Let $X_0, X_1, \ldots$ be a stationary Markov chain, and let a stored record M be generated from X_0, so that M is conditionally independent of the future trajectory $(X_1, X_2, \ldots)$ given X_0. If X_t is retained when M is erased, then the ideal Landauer content at age t is*

$$L_t = H(M \mid X_t) = H(M) - I(M; X_t). \tag{54}$$

Moreover, L_t is nondecreasing in t.

Proof. The identity is the definition of mutual information. Under the hypothesis the joint law factors as $p(x_0)\, p(m \mid x_0)\, p(x_t \mid x_0)\, p(x_{t+1} \mid x_t)$, from which $p(x_{t+1} \mid x_t, m) = p(x_{t+1} \mid x_t)$: the chain $M - X_t - X_{t+1}$ is Markov. Data processing then gives $I(M; X_{t+1}) \leq I(M; X_t)$, hence the monotonicity of L_t. □

The unconditional reset cost $H(M)$ has not changed. What changes is the share of that cost that can be reversibly discharged by correlation with the current state. An old record is less useful for predicting the present and, for exactly the same reason, harder to erase using the present as side information.

Example 1 (Symmetric binary Markov record)**.** Let X_t be a stationary fair bit that flips independently with probability $p \in (0, 1/2)$ per step, and store $M = X_0$. At age t,

$$q_t = \Pr\{X_t \neq X_0\} = \frac{1 - (1 - 2p)^t}{2}. \tag{55}$$

Therefore

$$I(X_0; X_t) = 1 - h_2(q_t), \tag{56}$$

$$L_t = H(X_0 \mid X_t) = h_2(q_t), \tag{57}$$

so predictive information and conditional reset content sum to one bit for every age. Figure 3 shows the exchange for $p = 0.05$.

The correlation law $(1-2p)^t$ often appears as a value-of-information decay. Equations (56)–(57) give it a second interpretation: age transfers one fixed bit from side-information-assisted reversibility into unavoidable reset uncertainty.

VIII. Relation to prior work

Landauer identified logical irreversibility with unavoidable heat generation in physical computation [1]; Bennett showed that arbitrary computation can be embedded in a logically reversible process, postponing the cost to the erasure of retained history [2]. Later work placed measurement and erasure in explicit thermodynamic accounting frameworks [9], [11]. Del Rio *et al.* showed that erasure work is governed by entropy conditioned on an observer's side information [3]. Faist *et al.* characterized the work cost of a general logical process through information discarded conditional on its output [4]. In an algorithmic, single-string setting, Zurek identified the thermodynamic cost of erasing a record with its compressed length [12], and Wolf sharpened the erasure principle to the compressibility of the string given the erasing agent's knowledge [13]. The present paper is a block-coding, lossy, and consumer-constrained counterpart of that observation: the object whose conditional compressibility is priced is a description that must remain decodable by a consumer lacking the side information available at reset. These results supply the physical accounting used here; this paper does not alter Landauer's principle (Remark 2).

On the source-coding side, Shannon's fidelity-constrained coding theorem introduced the rate–distortion function [14]. Wyner and Ziv characterized lossy coding when side information is available only at the decoder [15]. Steinberg imposed common reconstruction, requiring the reproduction to be determined without relying on decoder-only side information [6]. Equation (17) is exactly that common-reconstruction functional.

The distinction between codebook size and output entropy is also classical. Entropy-constrained vector quantization replaces a fixed-rate objective by the entropy of the quantizer output [7]; conditional entropy-constrained quantization exploits a retained context when coding a sequence of quantizer outputs [8]. Those works are an important precursor to the two-resource view. Theorem 1 differs in holding the full fixed description available to a consumer that lacks S while pricing the conditional entropy left for a separate reset mechanism that has S. Its thermodynamic content depends on that lifecycle distinction, not on the claim that output entropy had previously been ignored.

A second neighboring line relates thermodynamic efficiency to rate–distortion, relevance, and prediction. Still derived the rate–distortion objective from a least-effort principle for preparing a representation [16]. Still *et al.* then identified nonpredictive information as a lower bound on dissipation in driven systems [17]; subsequent work showed that retaining irrelevant memory limits thermodynamic efficiency and connects an information engine to the information bottleneck [18]. The present model prices a different stage of the cycle: conditional reset of an already available description. Relevance is specified by an arbitrary

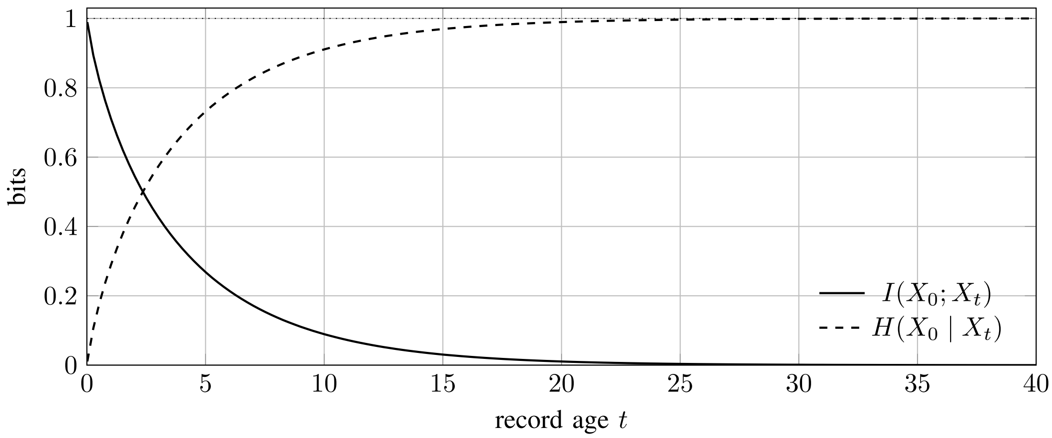

Fig. 3. For a symmetric binary Markov source with $p = 0.05$, predictive information lost with age is gained exactly as conditional Landauer content. The unconditional content of the stored fair bit remains one bit.

consumer distortion rather than only future prediction; the description must remain decodable without the reset side information; and the main object is a joint rate–work region rather than a single preparation or dissipation bound.

Observation Theory supplies the consumer geometry used in Section VI: a consumer pulls its output metric back to a positive semidefinite read operator, and the resulting distortion defines a consumer-relative rate–distortion problem [5]. The present paper starts after that geometry has been identified. It asks which parts of the resulting representation must be erased, under which retained correlations, and at what ideal work.

IX. Limits and extensions

Several extensions are natural, but they require a different physical or information-theoretic model.

Conditional entropy coding. The physical interpretation in this paper sits beside a substantial literature on entropy-constrained and conditional entropy-constrained quantization. The single-letter region above is stated for asymptotically long memoryless blocks; finite-dimensional quantizer structure, causal conditioning contexts, and implementable conditional reset circuits are separate design problems.

Finite blocklength and one-shot work. The proof uses typicality and asymptotic conditional entropy. At finite n, smooth max-entropies and error budgets replace Shannon entropy in the most general work bounds [4]. A second-order rate–work region would have two dispersions and need not be obtained by simply scaling a finite-blocklength rate–distortion formula.

Nondegenerate memories and finite time. If logical states have different energies, free-energy differences enter the accounting. If reset must finish in finite time or under time-symmetric controls, additional dissipation can dominate the Landauer term [10]. Proposition 5 allows different bath temperatures but retains degenerate logical memories and quasistatic reset.

Quantum side information. Conditional quantum entropy can be negative, allowing work extraction during erasure [3]. The rate coordinate would then have to be coupled to a quantum source-coding model. Nothing in Theorem 1 should be read as a quantum characterization.

Side information acquisition. The reset context S^n is assumed to survive for free. If it must itself be sensed, stored, transmitted, or erased, its lifecycle belongs in the optimization. A rate-limited erasure helper would produce a three-memory problem rather than the two-coordinate region studied here.

General Gaussian coupling. Proposition 4 assumes independently read and independently known modes. When Σ_X, $\Sigma_{X|S}$, and $P_{\mathcal{C}}$ are not simultaneously diagonalizable, the allocation program becomes a joint semidefinite optimization and the frontier need not separate across modes; its characterization is left open.

Sources with memory. Proposition 6 uses memory directly, but the coding theorem is i.i.d. Information-spectrum methods should extend the region to general sources, with conditional spectral information rates replacing single-letter mutual informations.

X. Conclusion

A useful observation is not only a message. It is a physical memory state with a lifecycle. Theorem 1 separates two costs of that state: the information a consumer must receive and the uncertainty a reset mechanism must irreversibly discard. Side information available only at reset can lower the second cost without helping the consumer with the first.

The distinction sharpens consumer-relative observation. The consumer determines which source distinctions must survive distortion. The reset context determines which of the stored distinctions remain predictable when the memory is cleared. The binary frontier shows why neither coordinate can be inferred from the other: a code that minimizes description capacity need not minimize reset work. Landauer's principle then prices the residue. In the exact case, that residue is $H(\mathcal{C}(X) \mid S)$. In the

Gaussian local model, it is the ordinary read-operator water-filling rate multiplied by $k_{\mathrm{B}}T\ln 2$. Across several consumers, it is the conditional entropy of their joint read rather than the sum of isolated records.

The materialization theorem states the practical boundary most plainly. Once a full volatile source record exists, later projection cannot make its erasure cheap. A consumer-relative observer reaches the lower work law only by avoiding that record, or by preserving enough reversible history to uncompute it. The thermodynamic consequence of Observation Theory is therefore not that irrelevant information disappears. It is that an observer should decide what is irrelevant before turning it into a bit that must later be erased.

Appendix A
Single-letter converse details

For completeness, we justify the last inequalities in (12) and (13). For the rate coordinate,

$$\begin{aligned} I(X^n;\widehat{X}^n) &= H(X^n) - H(X^n \mid \widehat{X}^n) \\ &= \sum_{i=1}^{n} H(X_i) - \sum_{i=1}^{n} H(X_i \mid X^{i-1}, \widehat{X}^n) \\ &\geq \sum_{i=1}^{n} \Big[H(X_i) - H(X_i \mid \widehat{X}_i) \Big] \\ &= \sum_{i=1}^{n} I(X_i;\widehat{X}_i). \end{aligned} \tag{58}$$

For the work coordinate,

$$\begin{aligned} I(X^n;\widehat{X}^n \mid S^n) &= H(X^n \mid S^n) - H(X^n \mid \widehat{X}^n, S^n) \\ &= \sum_{i=1}^{n} H(X_i \mid S_i) - \sum_{i=1}^{n} H(X_i \mid X^{i-1}, \widehat{X}^n, S^n) \\ &\geq \sum_{i=1}^{n} \Big[H(X_i \mid S_i) - H(X_i \mid \widehat{X}_i, S_i) \Big] \\ &= \sum_{i=1}^{n} I(X_i;\widehat{X}_i \mid S_i). \end{aligned} \tag{59}$$

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